IoT-ENABLED UPS SYSTEM WITH PREDICTIVE BATTERY DIAGONISTICS AND ADAPTIVE RUNTIME FORECASTING

A PROJECT REPORT

of

ECB1303 – EMBEDDED SYSTEM AND IOT DESIGN

Submitted by

MUTHULAKSHMI T	**(8115U24EC065)**
PRIYADHARSHINI E	**(8115U24EC079)**
SUBISHNA R	**(8115U24EC111)**
VARSHINI N	**(8115U24EC118)**

in partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

in

ELECTRONICS AND COMMUNICATION ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM, TRICHY

ANNA UNIVERSITY
CHENNAI 600 025

NOVEMBER 2026

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Under the Guidance of

Dr.R.SAMSON DANIEL, M.E., Ph.D

Department of Electronics and Communication Engineering

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
Under
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NOVEMBER 2026

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BONAFIDE CERTIFICATE

Certified that this project report titled "**IOT-ENABLED UPS SYSTEM WITH PREDICTIVE BATTERY DIAGONISTICS AND ADAPTIVE RUNTIME FORECASTING**" is the bonafide work of **MUTHULAKSHMI T(8115U24EC065), PRIYADHARSHINI E(8115U24EC079), SUBISHNA R(8115U24EC111), VARSHINI N(8115U24EC118)** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported here in does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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We declare that to the best of our knowledge the work reported herein has been composed solely by ourselves and that it has not been in whole or in part in any previous application for a degree.

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DEPARTMENT OF ECE

VISION

To be distinguished as a prominent program in Electronics and Communication Engineering Studies by preparing students for Industrial Competitiveness and Societal Challenges.

MISSION

M1. To equip the students with latest technical, analytical and practical knowledge

M2. To provide vibrant academic environment and Innovative Research culture

M3. To provide opportunities for students to get Industrial Skills and Internships to meet out the challenges of the society.

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PEO2: Graduates will provide innovative ideas and management skills to enhance the standards of the society by individual and with team works through the acquired Engineering knowledge.

PEO3: Graduates will be successful professionals through lifelong learning and contribute to the society technically and professionally.

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PO8: Individual and Collaborative Teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

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CO-PO MAPPING FOR PROJECT WORK

After successful completion of this course, the students should be able to

CO1: Analyze the internal architecture and modelling of an embedded processor using design tools.

CO2: Develop the embedded system networks using different protocols.

CO3: Identify the evolution of the Internet of Things (IoT) using internal components for networking.

CO4: Evaluate cloud and big data analytics using IoT Platform.

	POs											PSOs			
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	Average
CO1	2.60	1.40	2.00	1.80	1.80	3.00	1.40	1.80	1.40	2.00	3.00	1.80	1.40	0.80	1.87
CO2	2.60	1.20	2.00	1.20	2.00	3.00	1.20	1.40	1.60	1.80	3.00	1.80	1.80	0.20	1.77
CO3	2.60	1.20	2.00	1.60	1.00	3.00	1.20	1.60	1.80	1.20	3.00	1.80	1.80	0.80	1.76
CO4	2.60	1.20	1.20	2.00	1.20	3.00	1.20	1.80	1.60	1.20	3.00	1.80	1.80	1.20	1.77
CO5	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00
Average	2.68	1.60	2.04	1.92	1.80	3.00	1.60	1.92	1.88	1.84	3.00	2.04	1.96	1.20	2.03

SUSTAINABLE DEVELOPMENT GOALS (SDGs)

☐ Goal 1: No Poverty – End poverty in all forms everywhere

☐ Goal 2: Zero Hunger – End hunger and improve nutrition

☐ Goal 3: Good Health and Well-being – Ensure healthy lives

☐ Goal 4: Quality Education – Provide inclusive, equitable education

☐ Goal 5: Gender Equality – Empower women and girls

☐ Goal 6: Clean Water and Sanitation – Ensure safe water and sanitation

☑ Goal 7: Affordable and Clean Energy – Access to sustainable energy

☐ Goal 8: Decent Work and Economic Growth – Promote jobs and growth

☑ Goal 9: Industry, Innovation and Infrastructure – Build resilient infrastructure

☐ Goal 10: Reduced Inequalities – Reduce inequality

☐ Goal 11: Sustainable Cities and Communities – Make cities safe and sustainable

☑ Goal 12: Responsible Consumption and Production – Reduce waste

☑ Goal 13: Climate Action – Combat climate change

☐ Goal 14: Life Below Water – Protect oceans

☐ Goal 15: Life on Land – Protect ecosystems and biodiversity

☐ Goal 16: Peace, Justice and Strong Institutions – Promote justice and peace

☐ Goal 17: Partnerships for the Goals – Strengthen global partnerships

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ABSTRACT

IOT-ENABLED UPS SYSTEM WITH PREDICTIVE BATTERY DIAGONISTICS AND ADAPTIVE RUNTIME FORECASTING aims to develop an intelligent UPS monitoring system using an ESP32 microcontroller to estimate battery health, monitor load power consumption and predict backup time. A voltage sensor measures battery voltage, while ACS712 current sensors monitor individual load consumption. Battery health is estimated using a controlled pulse-load technique, where a MOSFET switches a power resistor to determine internal resistance based on the voltage drop and current change.

The ESP32 processes the sensor readings to calculate load power, estimate battery health and predict backup duration. It also recommends suitable load combinations based on the user's required backup time and predefined load priorities. Blynk IoT is integrated for remote monitoring of battery voltage, load consumption, estimated battery health and backup time. The proposed system combines battery diagnostics, load management and IoT-based monitoring to improve battery utilisation and ensure efficient UPS power management.

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CHAPTER-1

INTRODUCTION

1.1 INTRODUCTION

An Uninterruptible Power Supply (UPS) is an essential electrical device designed to provide continuous power to connected equipment during interruptions in the main power supply. It is widely used in homes, offices, hospitals, industries and communication systems where an uninterrupted power source is necessary. A UPS generally consists of a rechargeable battery, power conversion circuits and a control unit that manages the supply of electrical energy. During normal operation, the battery stores energy, which is supplied to the connected equipment whenever the main power source fails.

The battery is one of the most important components of a UPS, as its condition directly affects the reliability and duration of backup power. Over time, repeated charging and discharging cycles, operating temperature and ageing can cause the battery's performance to deteriorate. One of the important electrical characteristics associated with battery ageing is internal resistance. As the battery ages, its internal resistance generally increases, resulting in greater voltage drop under load and reduced power delivery capability. Therefore, regular assessment of battery condition is important to maintain reliable UPS operation.

Traditional UPS systems mainly focus on providing backup power and may offer only basic indications of battery status. They often provide limited information about the actual condition of the battery and the power consumed by individual connected appliances. In addition, users may find it difficult to estimate how long the available battery energy will support different combinations of loads during an extended power interruption. These limitations highlight the need for improved monitoring and diagnostic techniques.

Microcontrollers provide an effective solution for developing intelligent electrical monitoring systems. They can collect sensor data, process electrical parameters and perform calculations to support decision-making. The ESP32 is particularly suitable for such applications because it offers processing capabilities, multiple input and output interfaces and built-in Wi-Fi connectivity. Its communication capability also allows integration with IoT platforms for remote access to system information.

The integration of microcontroller-based diagnostics and IoT technology can improve the visibility and management of UPS performance. Monitoring battery characteristics and electrical load conditions helps users understand the operating status of the system and make informed decisions about power usage. Such an approach is useful in applications where maintaining essential electrical loads during unexpected power failures is important.

Thus, the development of an intelligent UPS monitoring and diagnostic system provides a practical approach to improving battery maintenance, power management and the reliability of backup supply.

1.2 EXISTING SYSTEM

Conventional Uninterruptible Power Supply (UPS) systems are designed to provide backup power to electrical equipment during interruptions in the main power supply. These systems generally consist of a rechargeable battery, charger circuit, inverter and control unit. Under normal operating conditions, the battery is charged through the charging circuit. When the main power supply fails, the inverter converts the stored DC energy from the battery into AC power to supply the connected loads.

In existing UPS systems, battery condition is commonly monitored using basic parameters such as terminal voltage, charging status and low-battery indications. Some advanced systems use battery management techniques to monitor voltage, temperature and charging or discharging current. These parameters provide general information about battery operation and help identify certain abnormal conditions. However,

terminal voltage alone may not accurately represent the actual condition of an ageing battery, as a battery can show a normal voltage while having reduced performance under load.

Battery internal resistance measurement is another method used in existing battery diagnostic systems. It helps assess battery condition by observing the relationship between voltage variation and current flow. Conventional testing methods may use dedicated battery analysers or impedance measurement instruments to evaluate internal resistance. Although these instruments provide useful diagnostic information, they may increase system cost and require additional testing equipment.

In conventional UPS installations, the connected loads are generally supplied together without detailed monitoring of individual appliance power consumption. Some systems display the total load level, but they may not provide separate information about the energy consumed by each connected load. Consequently, users may find it difficult to identify high-power-consuming appliances and manage the available battery energy effectively.

Backup-time estimation in many UPS systems is based on battery capacity and the connected load demand. However, actual backup duration can vary depending on battery ageing, load variations and operating conditions. In addition, traditional systems generally provide limited support for selecting suitable load combinations according to a required backup duration.

With the development of Internet of Things (IoT) technology, some modern UPS monitoring solutions provide remote access to operating parameters through cloud-based platforms. These systems improve accessibility and allow users to observe equipment status remotely. However, monitoring, battery diagnostics and intelligent load management may not always be integrated into a single system.

Therefore, existing UPS technologies provide essential backup power and basic monitoring capabilities, but limitations in battery condition assessment, individual load

monitoring and predictive power management create opportunities for further improvement.

1.3. DRAWBACKS FOUND IN EXISTING WORK

1. Conventional UPS systems mainly monitor battery voltage and charging status, which cannot accurately determine battery health.
2. Lack of internal resistance measurement makes early detection of battery ageing difficult.
3. Individual power consumption of connected loads is not monitored in basic UPS systems.
4. Backup-time estimation may be inaccurate as actual battery condition and load requirements are not considered.
5. No intelligent load recommendations based on required backup duration and appliance priority.
6. Limited IoT connectivity restricts remote monitoring of battery and load parameters.
7. Lack of automated load management reduces efficient utilisation of available battery power.

1.4. NEED FOR THE PROPOSED WORK

The reliability of a UPS mainly depends on the condition of its battery and the power consumed by the connected loads. In conventional UPS systems, battery monitoring is generally limited to voltage and charging status. These parameters alone cannot provide a clear indication of battery ageing or its actual performance under load conditions. Therefore, a system is required to assess battery health more effectively and provide useful information about its remaining backup capability.

The proposed work, IOT-ENABLED UPS SYSTEM WITH PREDICTIVE BATTERY DIAGONISTICS AND ADAPTIVE RUNTIME FORECASTING**,** aims to overcome these limitations by developing an intelligent UPS monitoring system using

an ESP32 microcontroller. A controlled pulse-load testing technique is used to estimate the internal resistance of the battery by observing the voltage drop when a known load is briefly connected. Since internal resistance generally increases as a battery ages, this measurement can help indicate battery deterioration and support battery health assessment.

Another important requirement is monitoring the power consumption of individual electrical loads. In the proposed system, three ACS712 current sensors are used to measure the current consumed by the fan, light and motor separately. The ESP32 processes the sensor readings to calculate individual and total power consumption. This helps users understand the energy requirement of each appliance and manage the available battery capacity more effectively.

The system also focuses on predicting the available backup time based on battery capacity, estimated battery health and the total power consumed by the connected loads. When the user enters a required backup duration, the system evaluates suitable combinations of loads and recommends which appliances can remain ON. The loads are considered according to their predefined priority, with the fan given the highest priority, followed by the light and motor. This feature helps users make better decisions during power interruptions.

Furthermore, Blynk IoT is integrated with the ESP32 to provide remote monitoring of important system parameters, including battery voltage, battery health estimation, individual load power and predicted backup time. Users can access this information through an IoT dashboard without being physically present near the UPS.

Thus, the proposed work is needed to combine battery internal resistance diagnosis, individual load monitoring, backup-time prediction, intelligent load recommendations and IoT-based monitoring in a single system. It aims to improve battery maintenance awareness, support efficient energy utilization and enhance the overall reliability of UPS operation.

1.5 APPLICATION OF THE PROPOSED SYSTEM

The proposed UPS monitoring system has various applications in areas where uninterrupted power supply and battery reliability are essential. It can be used in residential UPS systems to monitor battery health, estimate backup duration and manage connected household loads during power failures. In offices and commercial establishments, it helps maintain backup power for computers, networking devices and other essential equipment. Small-scale industries can use this system to monitor backup batteries and manage electrical loads efficiently during power interruptions.

The system can also be applied in telecommunication backup units, computer systems, laboratories and other installations that depend on battery-powered backup supplies. Its internal resistance testing feature provides an indication of battery ageing and helps identify possible battery deterioration at an early stage. The integration of Blynk IoT allows users to monitor important battery and load parameters remotely. Thus, the proposed system can support efficient battery utilisation, preventive maintenance and reliable power management in different applications.

CHAPTER-2

LITERATURE SURVEY

2.1. LITERATURE SURVEY

Continuous Battery Health Diagnosis by On-Line Internal Resistance Measuring – Jaime de la Peña Llerandi et al., 2019

Key Findings:

- Proposes an online battery health monitoring method based on internal resistance measurement.
- Uses voltage variations produced by the battery charger's current ripple to estimate internal resistance.
- Demonstrates the possibility of monitoring battery condition without disconnecting it from the UPS.
- Supports condition-based battery maintenance through continuous monitoring.

Challenges:

- Requires suitable measurement of voltage and current ripple.
- The accuracy of the method depends on the characteristics of the charger and battery.
- Integration with different UPS configurations may require additional calibration.

Estimation of Battery State of Health Using the Two-Pulse Method for LiFePO4 Batteries – Carolina Zuluaga et al., 2023

Key Findings:

- Investigates the two-pulse method for estimating battery State of Health (SOH).
- Uses voltage and current variations during controlled pulses to assess battery condition.

- Demonstrates the applicability of pulse-based testing for estimating battery SOH.
- Highlights the relationship between battery condition and its electrical response during pulse testing.

Challenges:

- Requires controlled current pulses and suitable measurement equipment.
- The test parameters must be selected according to the battery chemistry.
- Initial battery characterization is required for reliable estimation.

Determination of SoH of Lead-Acid Batteries by Electrochemical Impedance Spectroscopy – Monika Kwiecien et al., 2018

Key Findings:

- Studies the estimation of lead-acid battery health using Electrochemical Impedance Spectroscopy (EIS).
- Examines electrical impedance parameters to identify changes in battery condition.
- Explains the importance of internal resistance and impedance characteristics in battery health assessment.
- Demonstrates the use of impedance measurements for monitoring battery ageing.

Challenges:

- Requires specialized impedance measurement equipment.
- The testing and analysis process can be complex.
- Implementation may increase the overall cost of a simple battery monitoring system.

Model-Based State of Health Estimation of a Lead-Acid Battery Using Step-Response and Emulated In-Situ Vehicle Data – 2021

Key Findings:

- Presents model-based methods for estimating the State of Health of lead-acid batteries.
- Uses an equivalent circuit model to estimate battery capacity.
- Analyses battery response under changing operating conditions.
- Validates the proposed estimation methods using battery ageing data.

Challenges:

- Requires an accurate battery model and suitable parameter estimation.
- Battery ageing data are needed to validate the estimation process.
- Model development and implementation can increase computational complexity.

Rapid Estimation of Lithium-Ion Battery Capacity and Resistances from Short Duration Current Pulses – Journal of Power Sources, 2025

Key Findings:

- Investigates battery capacity and resistance estimation using short-duration current pulses.
- Uses voltage and capacity responses during pulse testing as indicators of battery health.
- Demonstrates rapid estimation using a lightweight neural network.
- Shows the possibility of implementing battery diagnosis on a microcontroller.

Challenges:

- The estimation method requires suitable battery data for training and validation.
- The proposed approach is based on lithium-ion battery datasets and requires adaptation for other battery chemistries.
- Accurate measurements are necessary during short-duration pulse testing.

BMS Control of UPS for Hybrid Energy System Using IoT – Arul Mozhi G. et al., 2025

Key Findings:

- Proposes an IoT-based battery management system for a UPS integrated with renewable energy sources.
- Uses an ESP32 microcontroller for monitoring battery parameters.
- Integrates Blynk IoT to display battery State of Charge (SOC) and State of Health (SOH).
- Provides remote monitoring of battery performance and supports maintenance decisions.

Challenges:

- Depends on network connectivity for remote monitoring.
- Focuses mainly on battery status and renewable energy integration.
- Further integration of individual load power measurement and intelligent backup-time-based load selection is required.

Battery Autonomy Estimation Method Applied to Lead–Acid Batteries in Uninterruptible Power Supplies – 2023

Key Findings:

- Proposes a method for estimating the remaining backup time of lead-acid batteries used in UPS systems.
- Develops an algorithm that learns battery behaviour from its operating conditions.
- Considers battery ageing and temperature variations during backup-time estimation.
- Provides a simplified approach that requires limited measurement data.
- Supports the prediction of available battery operating time during power interruptions.

Challenges:

- Requires proper battery characterization for reliable estimation.

- Changes in load conditions can affect prediction accuracy.
- Calibration may be necessary for different battery capacities and configurations.

A System for State-of-Health Diagnosis of Lead-Acid Batteries Integrated with a Battery Charger – Telles B. Lazzarin and Ivo Barbi, 2012

Key Findings:

- Presents a lead-acid battery charger with an integrated battery health diagnosis system.
- Uses internal impedance, DC voltage and operating temperature for battery condition assessment.
- Performs periodic measurements to identify changes in battery performance.
- Demonstrates the possibility of combining battery charging and health monitoring in a single system.

Challenges:

- Requires additional measurement and monitoring circuits.
- Periodic measurements must be analysed to identify battery deterioration.
- The system mainly focuses on battery diagnosis rather than individual load monitoring and backup-time-based load management.

Battery State Estimation for Lead-Acid Batteries under Float Charge Conditions by Impedance: Benchmark of Common Detection Methods – Julia Badeda et al., 2018

Key Findings:

- Compares different impedance measurement techniques for estimating the condition of lead-acid batteries.
- Studies valve-regulated lead-acid batteries under UPS float-charging conditions.
- Analyses the relationship between measured impedance and battery State of Health (SOH).

- Demonstrates that measurement frequency and excitation methods influence impedance readings.
- Highlights the importance of selecting suitable measurement techniques for reliable battery health estimation.

Challenges:

- Requires specialized impedance measurement instruments.
- Different measurement methods may produce different results.
- Accurate interpretation of impedance values requires suitable reference data and calibration.

Design and Development of a Battery State of Health Estimation Model for Efficient Battery Monitoring Systems – 2022

Key Findings:

- Develops an intelligent battery State of Health estimation model for battery monitoring systems.
- Uses battery parameters such as internal resistance, voltage and temperature for condition assessment.
- Investigates deep-learning-based methods for predicting battery health and ageing.
- Highlights the importance of continuous battery monitoring in UPS applications.
- Supports improved battery maintenance and early identification of performance deterioration.

Challenges:

- Requires sufficient battery data for model development and validation.
- Deep-learning techniques involve additional computational complexity.
- The model requires adaptation and testing for different battery types and operating conditions.

CHAPTER-3

PROJECT DESCRIPTION

3.1 PROPOSED SYSTEM DESCRIPTION

The proposed system, **“IOT-ENABLED UPS SYSTEM WITH PREDICTIVE BATTERY DIAGONISTICS AND ADAPTIVE RUNTIME FORECASTING,”** is designed to monitor battery health, estimate backup time and improve load management using an ESP32 microcontroller. A 12V, 8Ah sealed lead-acid battery is used as the power source. A voltage sensor monitors the battery voltage, while an ACS712 current sensor measures the total current consumed by the connected loads. A controlled pulse-load circuit consisting of a MOSFET and power resistor is used to estimate the battery's internal resistance by observing the voltage drop and corresponding change in current. The ESP32 processes these readings to calculate an approximate battery health index and assess the battery's condition.

The system also estimates the available backup time based on the usable battery energy and total load power consumption. Based on the user's required backup duration, it provides suitable load management recommendations to utilise the available battery energy efficiently. Relay modules are used to control the connected loads, while a 16×2 LCD displays important operating parameters such as battery voltage, load power, estimated battery health and backup time. Additionally, Blynk IoT is integrated to enable remote monitoring of system performance through a mobile dashboard. Thus, the proposed system combines battery health estimation, backup-time prediction and load management to provide an intelligent and cost-effective UPS monitoring solution.

3.2 MATHEMATICAL CALCULATIONS INVOLVED

1. Parameters selected for calculation

Parameter	Selected value
Nominal voltage	12 V
Battery capacity	8 Ah
Depth of Discharge (DoD)	0.5 (50%)
State of Health (SoH)	0.9 (90%)
System efficiency (η)	0.9(90%)

2. Approximate internal resistance and SoH

For illustration, assume:

- Reference internal resistance of a new battery = 0.025 Ω
- Estimated internal resistance = 0.0278 Ω

Using:

$$SoH = \frac{R_{new}}{R_{measured}} \times 100$$

$$SoH = \frac{0.025}{0.0278} \times 100$$

$$\boxed{SoH \approx 89.9\% \approx 90\%}$$

3. Available battery energy calculation

$$E_{usable} = V_{nominal} \times Ah \times DoD \times SoH \times \eta$$

$$E_{usable} = 12 \times 8 \times 0.5 \times 0.9 \times 0.9$$

$$E_{usable} = 38.88Wh$$

4. Backup –time calculation

$$Runtime = \frac{E_{usable}}{P_{total}}$$

$$Runtime = \frac{38.88}{20}$$

$$\boxed{Runtime = 1.944\ hours}$$

3.3 CIRCUIT DIAGRAM AND ITS EXPLANATION

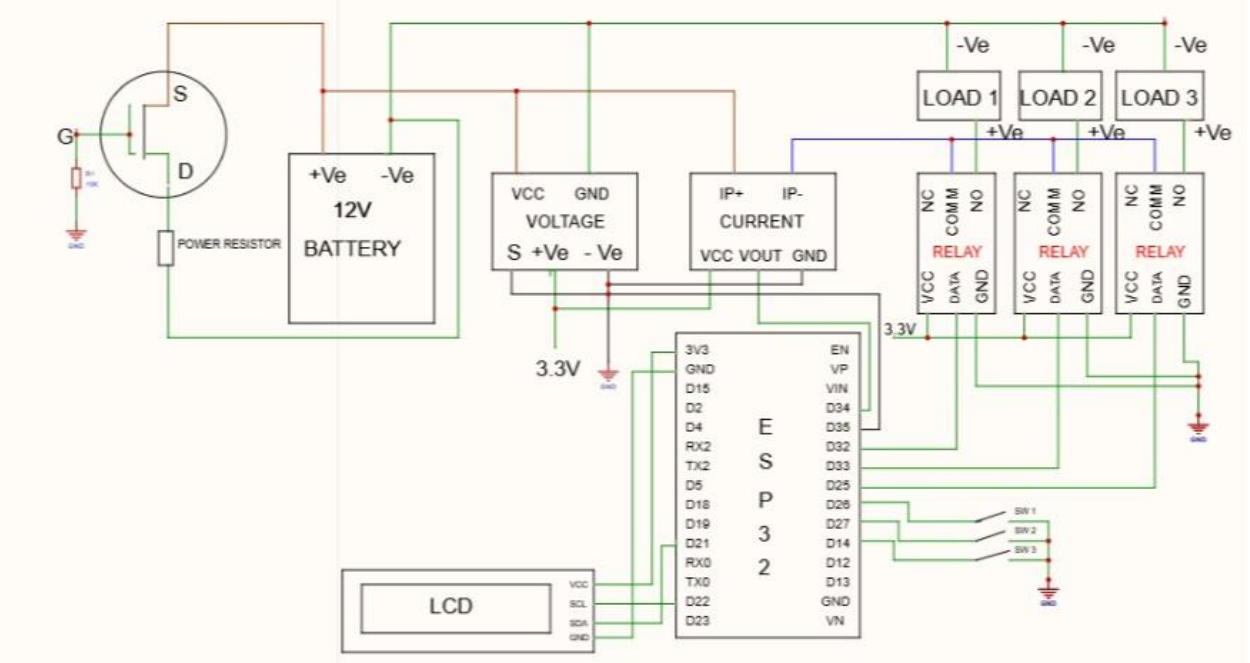


Figure 3.1: Circuit diagram of UPS Monitoring System

The circuit diagram illustrates the hardware arrangement of the proposed UPS monitoring system using an ESP32 microcontroller. A 12V battery acts as the main power source for the connected loads and monitoring circuit. The system consists of a voltage sensor, current sensor, MOSFET-based pulse-load circuit, relay modules, push-

button switches and an LCD display.

The voltage sensor is connected across the battery to measure its voltage continuously. The measured voltage is supplied to the ESP32 for monitoring and battery condition assessment. The current sensor is connected in the battery supply path to measure the current consumed by the connected loads. These voltage and current readings are processed by the ESP32 to calculate power consumption and estimate the available backup time.

A MOSFET and power resistor forms the internal resistance testing circuit. The MOSFET controls the connection of the power resistor to the battery for a short duration, producing a controlled load. The ESP32 uses the battery voltage readings before and during this test, along with the change in current, to estimate the battery's internal resistance. This value is used as an indicator of battery ageing and health.

Three relay modules are used to control the fan, light and motor independently. The ESP32 operates these relays according to the load requirements and the selected operating conditions. Three switches are provided as user inputs to control or select the corresponding loads. An LCD display is connected through the I2C interface to display important system information.

The ESP32 acts as the central processing unit, collecting sensor readings, controlling the relays and performing the required calculations. The processed parameters can also be transmitted to the Blynk IoT platform for remote monitoring. Thus, the circuit integrates battery monitoring, internal resistance testing, load control and backup-time prediction into a single system.

3.4 COMPONENTS USED

Table 3.1 Component List of the UPS Monitoring System

S. No.	Component	Specification	Quantity
1	ESP32 Development Board	ESP32 DevKit V1, 3.3V logic	1
2	Rechargeable Battery	12V, 8Ah sealed lead-acid battery	1
3	Voltage Sensor	0–25V DC voltage sensor module	1
4	Current Sensor	ACS712, 20A module	1
5	Relay Module	5V,1-channel relay module	3
6	MOSFET	IRLZ44N N-channel MOSFET	1
7	Power Resistor	15Ω, 10W	1
8	Gate Pull-down Resistor	10kΩ	1
9	LCD Display	16×2 LCD with I2C interface	1
10	Bulb	12V DC bulb	1
11	DC Fan	12V DC fan	1
12	Soldering Iron	12V DC soldering iron	1
13	Push Button Switches	Momentary push-button switches	3
14	Connecting Wires	Jumper wires	As required
15	PCB / Breadboard	Circuit assembly board	1

1. ESP32 Development Board

The ESP32 Development Board is the main processing unit of the proposed system. It receives voltage and current readings from the sensors, processes the collected data and performs the required mathematical calculations. It also controls the relay modules and communicates with the Blynk IoT platform through its built-in Wi-Fi facility. Its processing capability and multiple GPIO pins make it suitable for integrating different hardware modules.

2. 12V, 8Ah Lead-Acid Battery

The 12V, 8Ah sealed lead-acid battery acts as the primary power source for the UPS system. It supplies electrical energy to the connected loads during power interruptions. The battery voltage is continuously monitored to evaluate its operating condition. Its internal resistance is estimated using a controlled pulse-load technique to identify possible battery ageing and deterioration.

3. Voltage Sensor Module

The voltage sensor module is used to measure the battery voltage. It operates using

a resistive voltage-divider arrangement that scales the input voltage to a suitable level for measurement by the ESP32. The measured voltage is used for battery monitoring, internal resistance estimation and backup-time prediction. A 0–25V DC voltage sensor module is used for the 12V battery.

4. ACS712 Current Sensor

The ACS712 is a Hall-effect-based current sensor used to measure the current flowing through the connected electrical loads. It converts the measured current into a proportional analog voltage signal that can be read by the ESP32. The current measurement is used to calculate load power consumption and estimate the total power requirement of the system.

5. Relay Module

The 5V relay module acts as an electrically controlled switch for operating the connected loads. Three relay modules are used to control the bulb, DC fan and soldering iron independently. The ESP32 sends control signals to the relay modules to switch the respective loads ON or OFF according to the selected operating conditions and load recommendations.

6. IRLZ44N MOSFET

The IRLZ44N is an N-channel power MOSFET used as an electronic switching device in the internal resistance testing circuit. It controls the connection of the power resistor to the battery for a short duration. This produces a controlled load condition, allowing the ESP32 to compare the battery voltage before and during the test.

7. Power Resistor

A power resistor is connected in the pulse-load testing circuit along with the MOSFET. When the MOSFET is switched ON, the resistor draws a controlled current from the battery and produces a measurable voltage drop. This voltage drop, along with the change in current, is used to estimate the internal resistance of the battery.

8. 16×2 LCD Display with I2C Module

The 16×2 LCD display provides the user with important system information. It

displays parameters such as battery voltage, load power, estimated battery health and predicted backup time. The I2C interface reduces the number of GPIO connections required between the display and ESP32.

9. LM2596 Buck Converter

The LM2596 DC-DC buck converter is used to step down the 12V battery supply to a regulated 5V output. This provides a suitable power supply for the ESP32 development board and other compatible electronic modules. It helps maintain a stable supply voltage for the control circuit.

10. Push Button Switches

Three push-button switches are used as manual input devices. They allow the user to select or control the individual loads. The corresponding input signals are read by ESP32, which operates the relay modules according to the programme logic.

11. Bulb

The bulb is used as one of the electrical loads connected to the UPS system. It represents a lighting appliance and consumes electrical energy supplied by the battery. Its operation is controlled through a relay module, and its power consumption contributes to the total load calculation.

12. DC Fan

The DC fan is used as a cooling or ventilation load in the proposed system. It operates using the DC supply from the battery and is controlled independently through a relay module. Its current consumption is considered while calculating the total load power and available backup time.

13. Soldering Iron

The soldering iron is included as the third electrical load in the proposed system. It represents a comparatively higher-power appliance. Its operation is controlled through a relay module, and its power requirement is considered when estimating battery backup time and recommending suitable load combinations.

14. Connecting Wires and Breadboard

Connecting wires are used to establish electrical connections between the ESP32, sensors, relay modules and other circuit components. The breadboard is used for assembling and testing the electronic circuit during prototype development. Suitable wire thickness and secure connections are important for handling the load current safely.

3.5 SOFTWARE USED

1. Arduino IDE

Arduino IDE is used as the primary programming environment for the ESP32 microcontroller. The program is written and uploaded to the ESP32. It handles sensor readings, internal resistance estimation, load power calculations, backup-time prediction and relay control.

2. Wokwi Simulator

Wokwi is an online electronics simulator used to design and simulate the ESP32-based control circuit. It helps verify component interfacing, program execution, sensor inputs, LCD display operation and relay control before implementing the hardware prototype. It also helps identify errors in circuit connections and program logic.

3. Blynk IoT Platform

Blynk IoT is used for remote monitoring of the UPS system. It provides a dashboard to display battery voltage, load power consumption, estimated battery health and predicted backup time. The ESP32 sends the processed data to the Blynk platform through its built-in Wi-Fi connectivity.

4. Embedded C/C++ Programming

Embedded C/C++ is used to develop the control program for the proposed system. It processes sensor readings, performs mathematical calculations, controls the connected loads and updates the Blynk IoT dashboard.

5. ESP32 Board Support Package

The ESP32 board support package is installed in Arduino IDE to enable programming and uploading of code to the ESP32 development board. It provides the necessary board configuration and hardware support.

3.5 WORKING PRINCIPLE

The proposed system, **"**IOT-ENABLED UPS SYSTEM WITH PREDICTIVE BATTERY DIAGONISTICS AND ADAPTIVE RUNTIME FORECASTING**,"** operates using an ESP32 microcontroller to monitor battery voltage, estimate battery health, calculate load power consumption and predict the available backup time. The 12V, 8Ah lead-acid battery supplies power to the connected loads, while the voltage sensor continuously monitors the battery voltage. The ACS712 current sensor measures the current flowing through the common supply line. These sensor readings are given to the ESP32 for processing.

The battery's internal resistance is estimated using a controlled pulse-load technique. Initially, the battery's open-circuit voltage is recorded. A power resistor is then connected for a short duration through an IRLZ44N MOSFET to create a controlled load. The voltage drop and corresponding change in current are measured. The internal resistance is calculated using the voltage difference divided by the change in current. A higher internal resistance indicates battery ageing and reduced performance. Based on this value, the system calculates an approximate battery health index.

The system also monitors the power consumption of the connected loads, namely the bulb, DC fan and soldering iron. The ESP32 uses the current and voltage information to estimate power consumption and determine the total load demand. Based on the estimated usable battery energy and total power consumption, the available backup time is calculated. When the user enters the required backup duration, the controller evaluates suitable load combinations and provides recommendations to avoid excessive battery discharge. The relay modules are used to switch the loads according to the selected operation.

The calculated battery voltage, estimated health index, load power and predicted backup time are displayed on the 16×2 LCD. In addition, the Blynk IoT platform provides remote monitoring of the system parameters through an online dashboard. The LM2596 buck converter supplies the regulated voltage required by the ESP32 and other low-voltage components. Thus, the system combines battery health estimation, load monitoring and backup-time prediction to improve UPS performance and power management.

3.6 ADVANTAGES

1. Battery Health Monitoring: Estimates battery health by analysing its internal resistance using a controlled pulse-load technique.
2. Backup Time Prediction**:** Calculates the approximate backup time based on battery capacity and connected load power consumption.
3. Load Management: Suggests suitable load combinations according to the user's required backup duration.
4. Power Consumption Monitoring: Monitors the current consumption of the UPS load circuit.
5. Remote Monitoring: Blynk IoT enables users to monitor battery voltage, estimated health, load power and backup time remotely.
6. Real-Time Display: Displays important system parameters on a 16×2 LCD.
7. Improved Battery Utilization: Helps avoid unnecessary battery discharge by providing load management recommendations.
8. Low-Cost Implementation: Uses commonly available electronic components and an ESP32 microcontroller to provide an economical monitoring solution.

CHAPTER-4

RESULTS AND DISCUSSION

4.1 PROTOTYPE

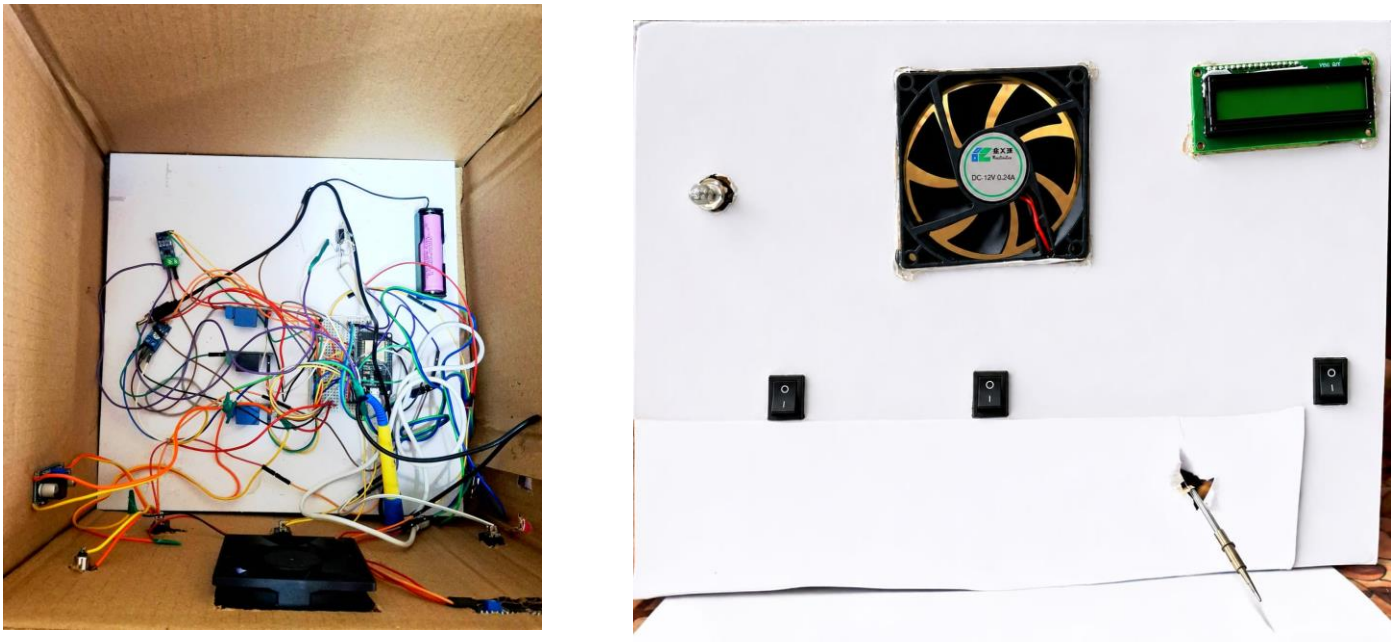


Figure 4.1: Prototype Kit of UPS Monitoring System

The prototype of the proposed UPS monitoring system is developed using an ESP32 microcontroller as the main processing and control unit. It integrates battery voltage monitoring, current sensing, internal resistance testing, load control and backup-time prediction in a single hardware setup. A 12V, 8Ah sealed lead-acid battery is used as the backup power source for the system.

The voltage sensor module is connected across the battery terminals to measure the battery voltage continuously. The ACS712 current sensor is connected in the common supply path to measure the total current drawn by the connected loads. These measured parameters are provided to the ESP32, which processes the readings and performs the required mathematical calculations.

For battery health estimation, a controlled pulse-load circuit consisting of an IRLZ44N MOSFET and a power resistor is incorporated. When the MOSFET is switched ON for a short duration, the resistor draws a controlled current from the battery.

The resulting voltage drop and change in current are used to estimate the internal resistance of the battery.

The estimated resistance is then compared with a reference value to obtain an approximate battery health index.

The prototype includes three relay modules to control the connected loads, namely a bulb, DC fan and soldering iron. Based on the load power consumption and the user's required backup duration, the ESP32 calculates the expected backup time and provides suitable load management recommendations. The LM2596 buck converter provides a regulated 5V supply for the required electronic components.

A 16×2 LCD is used to display the battery voltage, estimated battery health, load power and predicted backup time. In addition, the Blynk IoT platform is integrated to provide remote access to the monitored parameters through an online dashboard. The complete prototype demonstrates how microcontroller-based monitoring and internal resistance testing can support battery condition assessment and efficient UPS power management.

4.2 SIMULATION OUTPUT

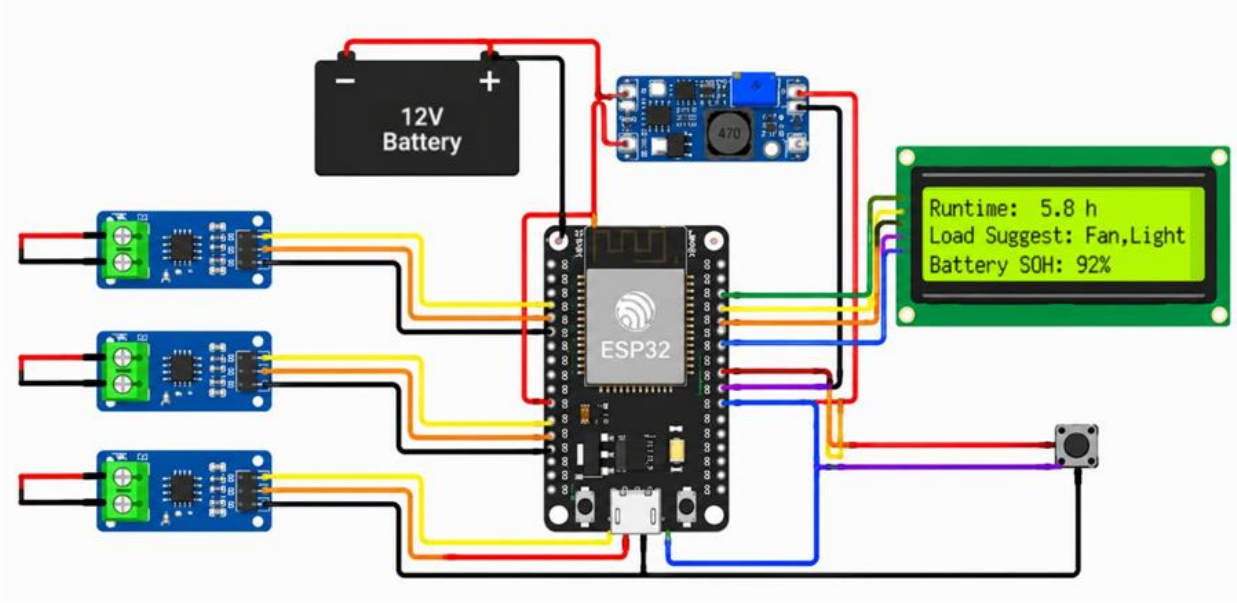


Figure 4.2: Wokwi Simulation Output of UPS Monitoring System

4.3 SIMULATION OUTPUT AND EXPLANATION

The proposed UPS monitoring system is simulated using the Wokwi online simulator to verify the circuit connections and program functionality before hardware implementation. The simulation consists of an ESP32 microcontroller, potentiometers, a 16×2 LCD display and other interfacing components.

In the simulation, potentiometers are used to provide variable analog input values to the ESP32. These adjustable inputs represent the changing load conditions and corresponding sensor readings. By varying the potentiometer positions, different input conditions can be created to test how the controller responds to changes in load demand, current and battery voltage. The ESP32 processes these input values and performs the required calculations for battery monitoring, health estimation and backup-time prediction. The LCD displays the calculated parameters based on the input values received by the microcontroller. When the potentiometer values are changed, the corresponding readings and calculated outputs are updated. This helps verify the operation of the program, the input-processing mechanism and the display functionality under different simulated conditions.

In the actual hardware implementation, the potentiometers used in the simulation are replaced by the corresponding physical components. The system uses a voltage sensor to monitor battery voltage and an ACS712 current sensor to measure the current flowing through the common supply line. Two physical loads are connected in the implemented setup to demonstrate load consumption and backup-time estimation. Thus, Wokwi simulation provides a preliminary platform for testing the control logic and calculations before implementing the system using actual hardware components.

CHAPTER-5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

The project **“**IOT-ENABLED UPS SYSTEM WITH PREDICTIVE BATTERY DIAGONISTICS AND ADAPTIVE RUNTIME FORECASTING**”** presents an intelligent approach to UPS battery monitoring using an ESP32 microcontroller. The system estimates battery health by measuring internal resistance through a controlled pulse-load technique and monitors battery voltage and load current. By processing these parameters, the system provides an approximate assessment of battery condition and helps identify performance deterioration. The implementation demonstrates how embedded technology can be used to improve conventional UPS monitoring systems.

Furthermore, the system predicts the available battery backup time based on usable battery energy and connected load power consumption. It also provides load management recommendations according to the user's required backup duration, helping to utilise battery energy efficiently. The integration of Blynk IoT enables remote monitoring, while the LCD provides local display of important system parameters. Overall, the proposed system offers a cost-effective solution for battery health estimation, backup-time prediction and efficient UPS power management, with scope for further improvement in measurement accuracy and automatic load control.

5.2 SDG MAPPING AND JUSTIFICATION

This project supports the United Nations Sustainable Development Goals (SDGs) by promoting efficient energy utilisation, technological innovation and responsible battery management. The project is primarily aligned with the following goals:

SDG 7 – Affordable and Clean Energy:

The proposed system promotes efficient utilisation of stored electrical energy by monitoring battery performance, predicting backup duration and recommending suitable load combinations. This helps minimise unnecessary power consumption and ensures better utilisation of available battery energy.

SDG 9 – Industry, Innovation and Infrastructure:

The project applies microcontroller-based technology and IoT monitoring to improve conventional UPS systems. Internal resistance testing, battery health estimation and backup-time prediction provide an improved approach to battery monitoring and power management.

SDG 12 – Responsible Consumption and Production:

By monitoring battery health and identifying changes in internal resistance, the system helps users understand battery ageing and make informed maintenance or replacement decisions. Efficient load management also supports responsible battery usage and may reduce premature battery replacement.

SDG 13 – Climate Action:

The project encourages energy conservation through improved battery utilisation and load management. By supporting longer battery service life and reducing unnecessary battery replacement, the system has the potential to minimise battery-related waste and its environmental impact.

Overall, the proposed system contributes to sustainable energy management through battery health monitoring, efficient power utilisation and the application of embedded and IoT technologies.

5.3 FUTURE SCOPE

The proposed UPS monitoring system can be further improved by incorporating advanced monitoring and prediction techniques to increase its accuracy, reliability and efficiency. The following enhancements can be considered in future developments:

1. Individual Load Monitoring: Separate current sensors can be installed for each connected load to measure individual current and power consumption accurately.
2. Improved Battery Health Prediction: Advanced battery diagnostic algorithms can be implemented to estimate the State of Health (SoH) more accurately by considering internal resistance, temperature, voltage and charging characteristics.

3. Machine Learning Integration**:** Machine learning algorithms can be introduced to analyse battery performance and predict battery ageing, backup duration and possible failures.
4. Automatic Load Management: The system can be enhanced to automatically disconnect non-essential loads when the available battery backup becomes insufficient.
5. Battery Temperature Monitoring: Temperature sensors can be added to monitor battery temperature and improve the accuracy of battery health estimation.
6. Advanced IoT Dashboard: The Blynk IoT platform can be further enhanced with historical data storage, performance graphs, low-battery notifications and maintenance alerts.
7. Battery Capacity Testing: A controlled discharge testing mechanism can be incorporated to validate the estimated battery health and determine the actual remaining battery capacity.
8. Multi-Battery Support: The system can be expanded to monitor multiple batteries in larger UPS installations and provide a centralized battery monitoring facility.

These improvements can transform the proposed prototype into a more accurate and intelligent UPS monitoring systems capable of supporting predictive maintenance, efficient energy utilization and reliable backup power management.

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